

HUI PENG JOHN YAO

Ladera Ranch, CA • US Citizen • Open to Remote / Nationwide

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EDUCATION

University of California, Los Angeles

- B.S. in Computer Science

Expected Graduation: June 2029 | GPA: 3.87

Thomas Jefferson High School for Science and Technology (TJHSST)

- High School Diploma

June 2025 | GPA: 4.55 (Weighted) | SAT: 1580

Relevant Coursework:

- Foundations of Computer Science, Data Structures & Algorithms, Artificial Intelligence, Mobile & Web Application Development, Linear Algebra, Probability & Statistics
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CERTIFICATIONS

- AWS Certified Cloud Practitioner (2025)
 - The Complete Full-Stack Web Development Bootcamp — Udemy (2025)
 - AI Agents: Automation & Business with LangChain & LLM Apps — Udemy (2025)
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SKILLS

- **Programming Languages:** Python, Java, JavaScript, C++, R, HTML, CSS, GDScript
 - **Frameworks & Libraries:** React, Node.js, Express, Tailwind, Bootstrap, PyTorch
 - **Cloud & Databases:** AWS (Lambda, S3, DynamoDB, IAM, API Gateway, CloudFront, Route 53, Certificate Manager), SQL, PostgreSQL
 - **Data Analysis & Statistics:** Python (NumPy, pandas), R, Statistical Testing (ANOVA, t-tests, chi-square), Data Visualization
 - **APIs, Tools & Platforms:** Git, REST APIs, OAuth 2.0, Linux, Alexa Skills Kit, Google Drive API, Flowise, Cursor AI, Slack
 - **Leadership & Collaboration:** Team Leadership, Cross-Functional Collaboration, Technical Communication
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PROJECTS

Fit 4 You (July 2025 – Present) | *React, JavaScript, Tailwind, Node.js, AWS, Flowise, Expo, OAuth, REST APIs*

- Leading a 6-member team building an AI-powered personal fitness web and mobile application.
- Designed and implemented a responsive React + Tailwind frontend. Integrated AI workflows via Flowise to generate custom fitness recommendations for users. Added cross-platform mobile support using Expo to extend access beyond the web.
- Built and deployed scalable backend services using Node.js and AWS (Lambda, DynamoDB, API Gateway) supporting authentication, data storage, and API management. Implemented OAuth-based authentication and role-based access control.
- Planned beta launch targeting ~1,000 UCLA users with a free/premium subscription roadmap.

Infinite Delve (June 2025 – Present) | *React, Tailwind, Node.js, AWS, PostgreSQL, REST APIs, Flowise*

- Designing an AI-driven game feature focused on procedural crafting and narrative generation.
- Built a full-stack cloud application on AWS with a React frontend and RESTful backend.
- Engineered backend services using DynamoDB and PostgreSQL to support structured game state and progression data.

Alexa-Powered Journal Assistant (Dec. 2024 – Feb. 2025) | *Python, AWS Lambda, Google Drive API, DynamoDB, Alexa Skills Kit, OAuth 2.0*

- Built an AWS Lambda function in Python to securely access and parse journal entries from Google Docs using OAuth 2.0, storing structured data in DynamoDB for efficient retrieval.
- Published a fully functional Alexa Skill, enabling hands-free journaling through voice commands.

Personal Website (Sept. 2024 – Nov. 2024) | *Git, React, AWS S3, CloudFront, Route 53, HTML, CSS, Express, Bootstrap*

- Designed and built a responsive personal website showcasing projects, research, and technical experience.
- Deployed static assets to AWS S3 and configured CloudFront + HTTPS using AWS Certificate Manager.

Unibound Consulting (May 2025 – Dec. 2025) | *React, Tailwind, Node.js, HTML, CSS, Express*

- Led a five-person team providing college consulting services, generating ~\$15,000 in revenue.
- Spearheaded front-end development for a client-facing consulting platform and website.
- Collaborated closely with teammates to define requirements, timelines, and feature priorities.

Published Research (Oct. 2023 – May 2024) | *R, Python, Statistical Analysis*

- Designed and distributed a psychological survey using *Brief-COPE* via Centiment (139+ participants).
- Performed statistical analysis in R and Python, including ANOVA, paired t-tests, and chi-squared tests.
- Published findings in the *Whitman Journal of Psychology*, identifying correlations between coping strategies and academic stress.

CS31 Projects (Sept. 2025 – Dec. 2025) | *C++*

- Built a zombie fighting board game using object-oriented programming in C++.
- Developed a robust text-processing program handling complex edge cases to improve user experience.

AI Fundamental Projects (Sept. 2022 – May 2023) | *Python, PyTorch*

- Created recurrent and feed-forward neural networks to analyze and predict image recognition datasets.
- Trained neural network models using PyTorch to evaluate model performance.

Arcane Chess (Sept. 2023 – May 2024) | *Personal Project | Godot, GDScript*

- Developed a desktop game that combines the classic gameplay of chess with a card and mana system.
 - Implemented game logic, state management, and turn-based interactions in Godot.
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LEADERSHIP

Development Team Officer (Nov 2025 – Present) | *GLITCH @ UCLA*

- Led development of an authentication portal and frontend to support streamlined access for a large-scale hackathon.
- Designed and implemented OAuth-based authentication flows and integrated backend services using API Gateway.
- Coordinated with teammates to define requirements, timelines, and technical responsibilities.

Woodwind Captain (Sep 2021 – May 2025) | *TJHSST Marching Band*

- Led and mentored 40+ musicians; developed lesson plans and onboarding guides for new members.
- Organized sectionals and team-building activities, including custom card games, to foster collaboration and cohesion.

Science Olympiad Team Lead (Sep 2021 – May 2024)

- Led and competed in engineering and research events, including Experimental Design and Thermodynamics.
- Earned 20+ medals at regional, state, and national competitions through collaborative problem-solving.